Recovering and Counting Grapes Behind Foliage from a Single Moving Camera

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Abstract

Vineyard growers could benefit from a simple way to estimate crop without manually inspecting every vine, but leaves hide fruit from any single view. We investigate whether camera motion alone can recover enough hidden fruit to improve a downstream task: counting grapes. Using one phone on a laboratory rail as a surrogate for a vehicle-mounted camera, we align views on fruit with SIFT and RANSAC, then composite only colours observed in multiple views. Pixel recovery alone does not show that a reconstruction is useful, so we evaluate it twice: pixel-wise against a leaf-free photograph, and by count, using GrapeSAM, an independently published point model trained on the ViViD-5K vineyard dataset and left untuned on our data, against 25 hand-labelled grapes. At 20 mm spacing, fruit pixel recall rose from 42% to 87% with five foreground branches, and count F1 rose from 0.42 to 0.77. Benefits fell with sparser sampling and heavier occlusion: at six branches, 20 mm spacing gave only partial recovery (25% to 42% recall), with 14 of 27 views aligned, and 60 mm fell below the single view. Because all conditions share one physical scene, these results are a controlled proof of concept, not estimates of field accuracy or yield.

1. Introduction

The motivation for this project is a potentially accessible vineyard monitoring tool: a farmer could attach an ordinary smartphone to a tractor or other vehicle, record video while traversing vine rows, and ultimately obtain an estimate of the relative amounts of ripe and unripe grapes. Such a workflow using only a camera could avoid dedicated depth sensors, LiDAR, external positioning hardware and elaborate multi-camera installations. It would also make repeated inspection of the same rows more practical than manually examining each obscured cluster. This is the *intended application*, not a claim that a system mounted on a tractor has been field validated.

Occlusion is the principal vision challenge. Leaves, branches and neighbouring bunches conceal grapes, so an otherwise accurate single image segmenter can underestimate the visible crop. The value of additional viewpoints is illustrated by Hemming *et al*. [3]: in a sweet pepper study, no single camera position exceeded 69% detectability under a criterion requiring at least half of each fruit to be visible, whereas five viewpoints together achieved 90%. Their measure differs from ours, but demonstrates why collecting more views may be valuable. For a side facing phone travelling down a vineyard row, successive video frames naturally provide changing perspectives. Parallax causes nearby leaves and deeper fruit to move differently across images, potentially revealing berries that were hidden in the first frame.

We therefore investigate the *vision feasibility* of this deployment model using one smartphone and a repeatable horizontal rail as a laboratory analogue of vehicle movement. The objectives are to (i) demonstrate that multiple views from a single camera recover hidden grape regions, (ii) develop a registration and fusion pipeline that does not need measured scene depth or an external motion sensor, (iii) evaluate sensitivity to viewpoint spacing and occlusion, (iv) test whether recovered fruit improves a downstream task, by counting grapes with an independently trained model in addition to pixel-wise evaluation, and (v) explore red/green segmentation as an initial component of future ripe/unripe crop quantification. The present contribution is a controlled proof of concept, not a validated estimate of fruit mass, true maturity or harvest yield.

2. Related Work

2.1. Agricultural visibility and yield

Agricultural robots often address occlusion through active perception: deliberately moving cameras or manipulators to expose fruit otherwise hidden by crop structure. For routine vineyard monitoring, however, useful viewpoints may already be obtained passively as an existing farm vehicle travels along a row. Hemming *et al*. [3] quantified the benefit of multiple viewpoints for sweet peppers. For vineyard

applications, Millan *et al*. [5] instead used segmentation and a Boolean model to compensate statistically for partly occluded berries, obtaining $R^2 = 0.81$ on manually acquired vine imagery and $R^2 = 0.78$ for on the go three vine segments. Such approaches directly estimate yield, but depend on a model of hidden berries and an adequate segmentation mask. Our method addresses an earlier stage: increasing the fruit evidence available to a later red/green crop count or yield model using an ordinary moving phone rather than additional sensing hardware. Because our results evaluate visible target recovery rather than harvested weight, comparisons with these yield accuracies must remain conceptual.

2.2. Synthetic apertures and learned de-occlusion

Figure 1 illustrates the reconstruction strategies compared by Vaish *et al*. [7]. Synthetic aperture imaging combines samples acquired from distinct camera positions after refocusing them onto a selected depth. Vaish *et al*. [7] studied stereo, synthetic aperture focus, and robust statistics such as colour medians for occluded surface reconstruction. Foreground objects that do not lie at the focal plane become spatially inconsistent across aligned views, whereas targets at that plane reinforce. However, the refocusing depth matters, and an entirely hidden object cannot be recovered by merely averaging photographs. More recent light field work, such as DeOccNet [8], learns occlusion free views from angularly sampled input images, at the cost of training requirements and a different sensor or data collection regime. We use the translation rail only to validate camera motion principles in the laboratory. The intended vineyard system would obtain image diversity from the motion of an existing vehicle and use explicit geometric and photometric consistency rather than a dedicated light field sensor.

2.3. Feature matching and semantic masks

SIFT produces distinctive image descriptors designed to tolerate rotation and scale changes [4]. RANSAC robustly fits geometric relationships when tentative correspondences contain outliers [2]. Together they provide a practical registration mechanism even when some fruit features disappear behind foliage. Unlike a physical depth refocusing model, feature derived transforms can align corresponding target observations without requiring the user to supply their depth. This is conditional on sufficient unoccluded correspondences and an adequate transformation model. For the red/green extension, SAM 3 provides promptable concept segmentation of objects [1]. Its semantic prior is potentially useful when green fruit and leaves have overlapping colour distributions, but model output still requires validation on the target domain.

2.4. Grape counting

Berry-level grape datasets are uncommon. ViViD-5K [6] provides 5,000 vineyard images with over 648,000 berry centroids across 13 varieties, along with GrapeSAM, a point localiser followed by SAM, reporting a counting MAE of 21.46 berries per image on its test set. Its images average about 130 berries, so that figure is not comparable with our 25 grape scene. We use its published point model as an independent counter, not as a method we extend. Its authors note reduced performance on indoor and uniform backgrounds, which applies to our setting.

3. Capture System and Design Evolution

3.1. Rail mounted image acquisition

To isolate the vision problem before mounting a phone on a moving vehicle, the project uses an indoor scene containing red grapes on branches and removable foreground foliage. Early demonstrations used toy fruit and pipe cleaner leaves. Later trials included denser leaf and branch arrangements and green grapes. A phone camera views this arrangement through a rectangular enclosure. The camera is moved horizontally on a rigid rail while aiming toward a nominally stationary target scene. Illumination was kept approximately controlled and static during a sequence, reducing temporal appearance changes not attributable to parallax. Figure 2 shows this experimental apparatus. The rail is a controlled motion surrogate, not equipment a farmer would need to install.

The evaluated scans sample the rail at 20 mm intervals for 28–30 images. Wider spacings of 40 and 60 mm are formed from subsets of those views. Unlike an array of synchronised cameras, this arrangement needs only one RGB camera. The rail provides known test positions, while the final feature based method registers image content rather than reading an encoder or depth sensor. However, sequential capture is vulnerable to moving leaves, illumination changes and camera rotation.

3.2. Envisaged vineyard use: RGB capture during vehicle travel

The proposed field workflow is intentionally simple. A side facing smartphone, secured to an existing tractor, utility vehicle or other row traversing machine, would capture a sequence of overlapping RGB frames of the vine canopy. Software would select neighbouring views, register shared fruit features directly from the images, suppress changing foreground occlusions, then segment recovered grapes into red and green appearance categories. Aggregated observations could ultimately give growers an estimate of ripe versus unripe crop along inspected rows. Crucially, the proposed *feature based* route does not require an external GPS receiver, wheel encoder, depth sensor or measured camera



Figure 1. Examples of reconstruction behind plants using entropy, median, focus and stereo, from Vaish *et al*. [7] (top row of their Fig. 4, reproduced in the team's slides). This prior work illustrates the sensitivity of different synthetic aperture reconstruction methods to foliage. These images are not results from our experiment.

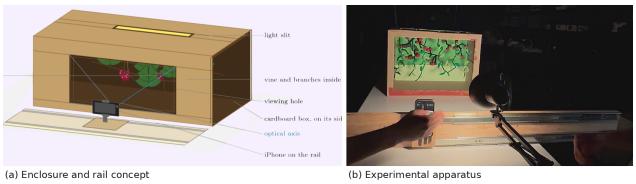


Figure 2. Controlled proof of concept capture rig from the team's presentation: (a) enclosure with a horizontal camera rail and separated fruit/foliage layers, and (b) the actual phone, rail, lighting and target. The rail reproduces camera translation for benchmarking. A future vineyard implementation would instead use a phone carried by a moving tractor or similar vehicle.



Figure 3. Original slide images from the depth guided synthetic aperture trial: densely occluded grape scene (left) and fused refocused view (right). Some foliage is suppressed, but substantial foreground remains. The output is not a segmented fruit mask.

step size: the relative image alignment is estimated visually. A mounting bracket and computation would still be necessary. No complete implementation on a vehicle is claimed here. The laboratory rail supplies controlled spacing for evaluation, not a requirement of the eventual system for farmers.

3.3. First approach: depth guided synthetic aperture

For a static 3D point imaged by cameras translated parallel to their image plane, the simplified pinhole model gives the horizontal image shift relative to a chosen reference view:

$$\Delta u_i \simeq \frac{f_x B_i}{Z_t}, \tag{1}$$

where f_x is horizontal focal length in pixels, B_i is horizontal camera translation from the reference, and Z_t is target depth along the camera optical axis (both B_i and Z_t in the same physical units). Under the assumed fixed camera orientation, images are shifted so points at Z_t agree. A foreground leaf at $Z_o < Z_t$ shifts by a different amount, with relative parallax approximately $f_x B_i(1/Z_o - 1/Z_t)$.

The prototype fused target aligned images by per pixel median or by a masked average:

$$I_{\text{fused}}(\mathbf{x}) = \frac{\sum_i V_i(\mathbf{x})\, I_i^{\text{warp}}(\mathbf{x})}{\sum_i V_i(\mathbf{x}) + \epsilon}, \tag{2}$$

where V_i indicates usable pixels and ϵ avoids division by zero. The median can suppress inconsistent foreground pixels when enough views reveal the background, whereas a mask excludes pixels believed to belong to occluders. Both strategies suppress some foliage in simple scenes (Fig. 3). They also exhibit ghosting, residual foliage and visible target blurring when the assumed depth is inaccurate, a target spans different depths, masking is unreliable, or the target is visible too infrequently. Crucially, averaging does not produce a fruit only segmentation. These limitations motivated the final feature based approach.

4. Final Feature Based Reconstruction

4.1. Candidate grape segmentation

Let $I_i : \Omega_i \to \mathbb{R}^3$ denote RGB image i and $i = 1, \ldots, N$ the camera position. For each image, a binary fruit mask $C_i(\mathbf{x}) \in \{0, 1\}$ establishes which warped observations may contribute fruit pixels. The main configuration uses SAM 3 text-prompted fruit masks [1], with foliage masks excluding occluded pixels. A secondary configuration thresholds colour in Lab and RGB within a fixed image region. Colour is a limited appearance model: a shadowed or green grape may resemble leaves, and a reddish non-grape object may produce a false positive.

4.2. SIFT and RANSAC registration

SIFT keypoints and descriptors are extracted from candidate target regions and matched across overlapping viewpoints [4], using the OpenCV library. Each view is compared with its next four neighbours, keeping reciprocal matches passing a 0.78 ratio test. A quarter of matches are held out, and the remainder are tested by RANSAC, which repeatedly samples correspondences, estimates a geometric transform, and rejects matches inconsistent with that model [2]. For an accepted image pair (i, j), inlier correspondences satisfy

$$\|\mathbf{p}_j - \mathcal{T}_{ij}(\mathbf{p}_i)\|_2 < \tau_g, \tag{3}$$

where $\mathcal{T}_{ij}$ is a 2D similarity transform and $\tau_g = 1.5$ pixels. Pairs need at least 12 inliers, scale within 6% of unity and rotation under 3°. Accepted pairs are then solved jointly by iteratively reweighted least squares for one similarity $\mathcal{T}_{i\to r}$ per view, with the best-connected view as reference. Any view whose median held-out error exceeds 3 pixels is dropped. The final approach thus replaces a user provided absolute fruit depth or separately measured camera translation with correspondences observed directly in the frames, supporting the smartphone only deployment concept. It does *not* remove the need for image overlap, stable features or sufficiently limited camera motion. A single 2D mapping may also be inadequate for strongly three dimensional grape clusters.

4.3. Visibility aware multi-view fusion

Each source image and candidate mask is warped into the common reference domain:

$$\tilde{I}_i(\mathbf{x}) = I_i\big(\mathcal{T}_{i\to r}^{-1}(\mathbf{x})\big)\,, \quad \tilde{C}_i(\mathbf{x}) = C_i\big(\mathcal{T}_{i\to r}^{-1}(\mathbf{x})\big)\,. \tag{4}$$

Define $s(\mathbf{x}) = \sum_i \tilde{C}_i(\mathbf{x})V_i(\mathbf{x})$ as the number of geometrically valid, grape candidate observations at a reference pixel. At each pixel, fusion takes the median colour of candidate observations and keeps those within an RMS distance of 30 of it. The output pixel is observed when

$$\hat{G}(\mathbf{x}) = \mathbb{1}[s(\mathbf{x}) \geq m]\,\mathbb{1}[B(\mathbf{x}) \geq B_{\min}]\,, \tag{5}$$

where s here counts agreeing views, $m = 3$, B is the rail span of those views and $B_{\min} = 40$ mm. The colour is copied from the single best agreeing view, not blended, so no colours are synthesised. Figure 4 illustrates this sequence using intermediate outputs from an earlier colour-mask run.

This design favours conservative, observation backed recovery. It can restore a region hidden in the reference photograph when other rail positions expose that region. It cannot reconstruct a berry hidden from *every* acquired view, and an overly stringent support rule may remove legitimate fruit observed only once. At the opposite extreme, permissive support or incorrect alignment can retain foreground pixels or introduce duplicated fruit. Registration graph connectivity is another failure mode: if a target lacks sufficiently reliable shared features across viewpoints, no consistent reference reconstruction can be formed. These effects are central to interpreting the severe occlusion experiments.

4.4. Registration graph and error propagation

Accepted RANSAC image pairs can be regarded as edges in a graph of viewpoints. Only views connected to the reference can contribute to its reconstruction. With SAM 3 masks at 20 mm spacing, between 84 and 95 of 102 tested pairs passed for three to five branches, but only 40 of 98 at six branches, where 13 views were dropped. More images therefore do not necessarily imply a larger usable aperture.

In the depth guided prototype, first order shift error from an incorrect fruit depth obeys $\delta(\Delta u) \simeq -(f_x B/Z_t^2)\delta Z_t$: sensitivity grows with baseline. For feature based fusion, incorrect image transforms likewise misalign true fruit, potentially rejecting it during colour consensus or generating duplicates. The supporting view threshold therefore balances recovery of partly seen fruit against false additions.

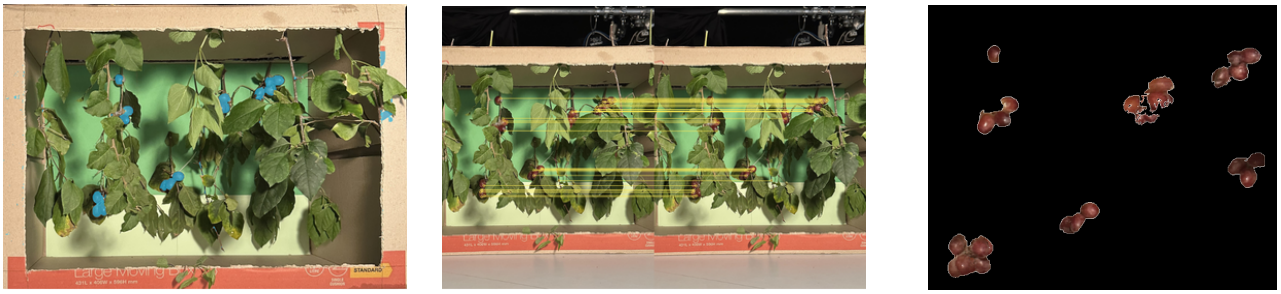
4.5. Red/green crop extension

The initial colour front end primarily selects red and purple fruit. Green grapes are difficult to separate from green foliage by colour alone. The team demonstrated a supplementary SAM 3 masking stage that identifies red and green fruit like regions and applies a reported red/green colour ratio threshold of 1.04 to distinguish the two categories (Fig. 5) [1]. This extension differs from the original Week 9 classical thresholding proposal. It targets the application level distinction between ripe coloured and unripe coloured grapes, rather than merely removing occlusion. A future system could aggregate segmented red and green instances per row to estimate relative crop amounts. One proposed farmer facing output is the ripe fraction $\hat{p}_r = \hat{N}_r/(\hat{N}_r + \hat{N}_u)$, where $\hat{N}_r$ and $\hat{N}_u$ are distinct grape counts after cross view deduplication. This metric was not measured here, and interpreting red/green appearance as ripeness requires variety specific validation. The current demonstration is *not* a validated physiological maturity, disease or health classifier. Neither the exact ratio definition nor an independent mask precision/recall experiment was supplied. We therefore report the threshold without inventing its channel formula or numerical classification accuracy.

5. Experimental Protocol

5.1. Factors and reference measurements

The evaluation varies *viewpoint spacing* (20, 40 and 60 mm) and *occlusion severity* (3, 4, 5 or 6 foreground branches).



(a) Colour candidates (b) Feature matching (c) Fused grapes

Figure 4. Final feature based processing stages using individual images from the presentation: colour candidate selection, SIFT/RANSAC correspondences and fused grape extraction. The extracted grapes are shown against black instead of the white or transparent slide background. Correspondence lines illustrate the registration stage, not its measured accuracy.



Figure 5. Original presentation images for the SAM 3 red/green extension, showing the mixed colour scene (left), binary fruit mask (centre) and red/green candidate overlay (right). The reported colour ratio threshold was 1.04. These are illustrative masks, not a validated ripeness classifier or a measured segmentation accuracy.

Each branch count has one rail capture of 28–30 views at 20 mm steps. Sparser spacings keep every second or third view around the same reference, so spacing changes the number of views as well as the baseline. A leaf-free photograph of the same fruit serves as pixel truth, and its grapes were hand labelled to give 25 berry centres. The photograph was taken from a different position, so it is registered to each capture's reference with a similarity transform fitted to features near fruit. We reviewed an overlay for every branch count; agreement is good overall, though some clusters show visible offset at five and six branches, where fruit lies at differing depths. All conditions share one physical arrangement of fruit, with no repeats, so the results describe these conditions and not a population.

All runs used scale 0.8 of the original resolution and the SAM 3 fruit and foliage masks stored with each capture. A secondary run replaced fruit masks with the red colour rule at 20 mm.

5.2. Pixel-wise evaluation

Given the registered truth fruit mask G and reconstructed mask $\hat{G}$, we report

$$R = \frac{|G \cap \hat{G}|}{|G|}, \quad P = \frac{|G \cap \hat{G}|}{|\hat{G}|}, \quad \text{IoU} = \frac{|G \cap \hat{G}|}{|G \cup \hat{G}|}. \quad (6)$$

The single view baseline applies the same formulas to the reference view's own fruit mask. Truth masks also come from SAM 3, so this evaluation shares a model with the reconstruction and is not fully independent.

5.3. Count-based evaluation

Pixel overlap can reward regions that are not separable into individual fruit, so we also ask how well an independent counter performs on each image. We use the point model from GrapeSAM [6], a VGG19-based localiser trained on the ViViD-5K vineyard dataset, with the authors' published weights and post-processing (confidence 0.05). None of its settings were tuned on our data, and it shares no training with our segmentation. It runs on three inputs: the occluded reference view, the reconstruction, and the registered leaf-free photograph, the last two restricted to fruit pixels on a black background so that all three are comparable.

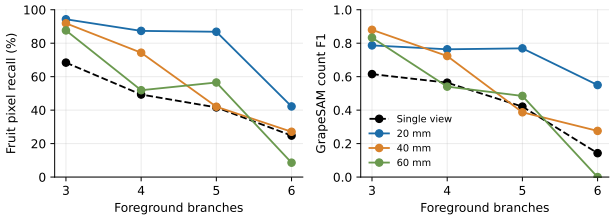


Figure 6. Fruit pixel recall (left) and GrapeSAM count F1 (right) against branch count. Dashed lines show the occluded single view. Spacing of 20 mm leads at four or more branches, while wider spacing can match or exceed it at three.

Detections are matched one to one to the 25 labels, closest first, within 24 pixels, and we report F1 and count. The tolerance is about the radius of one grape, so a match shows the detection falls on the grape, not that it is centred. We label the leaf-free run as a reference and not a strict upper limit, for reasons given in Sec. 6.2. Two secondary counters, SAM 3 instances and Hough circles, had settings selected on the four-branch capture using these same labels, so they are not independent of the labels.

5.4. Reproducibility

The experiment, tables and figures are generated by `run_experiments.py`, `analyse_experiments.py` and `make_report_assets.py`. As a check on the code, a repeat run on one capture reproduced its earlier reference view, 101 of 102 accepted pairs and 13,395 observed pixels.

6. Results

6.1. Pixel-wise recovery

Table 1 and Fig. 6 give fruit recovery by condition. At 20 mm spacing, recall rose from 68% to 94% with three branches, 49% to 87% with four, 42% to 87% with five, and 25% to 42% with six. Gains shrink as spacing widens, especially under heavy occlusion. The six branch, 60 mm run fell below the single view (25% to 9%).

Spacing effects are tied to alignment. Runs that lost views to validation recovered less, such as five branches at 40 mm, with 8 of 14 views kept and no gain in recall. This explains why 40 mm scored below 60 mm at five branches, where all 9 views survived, so sparser sampling does not degrade smoothly. Six branches at 20 mm is only a partial recovery: 14 of 27 views aligned, whole clusters are absent, and a strip of non-fruit passed through the masks.

6.2. Count-based results

Table 2 shows that the reconstruction improves counting at 20 mm: F1 rose from 0.62 to 0.79 with three branches, 0.56 to 0.76 with four, 0.42 to 0.77 with five, and 0.14 to 0.55 with six. These gains are from a counter that was neither trained nor tuned on our data. The picture is not uniformly positive. Wider spacing could remove the benefit: F1 was 0.54 against a single view 0.56 at four branches and 60 mm, 0.39 against 0.42 at five branches and 40 mm, and 0.00 at six branches and 60 mm, where no grape was detected. Totals can also mislead, since the three branch 20 mm reconstruction gave 36 detections for 25 grapes, while three branches at 40 mm gave exactly 25 with F1 0.88. We therefore treat F1 as the primary score.

Table 1. Pixel-wise results with SAM 3 masks. Single and Restored are fruit pixel recall (%) for the occluded reference view and the reconstruction. Views are those kept after alignment validation. IoU is for the reconstruction.

Branches	mm	Views	Single	Restored	IoU
3	20	28/28	68	94	0.76
3	40	14/14	68	92	0.80
3	60	9/9	68	88	0.80
4	20	28/28	49	87	0.72
4	40	14/14	49	74	0.65
4	60	7/9	49	52	0.49
5	20	28/28	42	87	0.73
5	40	8/14	42	42	0.40
5	60	9/9	42	57	0.54
6	20	14/27	25	42	0.38
6	40	8/13	25	27	0.26
6	60	3/9	25	9	0.08

The leaf-free photograph is a useful reference but not a strict ceiling. GrapeSAM overcounted it, with 29–40 detections for 25 grapes in the fruit-only runs, and three branch reconstructions exceeded its F1. As the same photograph scored between 0.74 and 0.81 under different registrations, differences of that size between conditions should not be interpreted. Figure 7 shows the three inputs for the five branch case.

6.3. Mask source and secondary counters

Colour masks were weaker than SAM 3 masks at 20 mm. Recall was 81%, 73% and 50% at three, four and five branches, against 94%, 87% and 87%, and F1 was 0.77, 0.70 and 0.32. At six branches the colour run failed, because fewer than three views passed alignment validation. Table 3 compares counters on the 20 mm reconstructions. SAM 3 instances scored highest, but their settings were selected using our labels and they share a model with the masks, so the GrapeSAM results remain the independent measure.

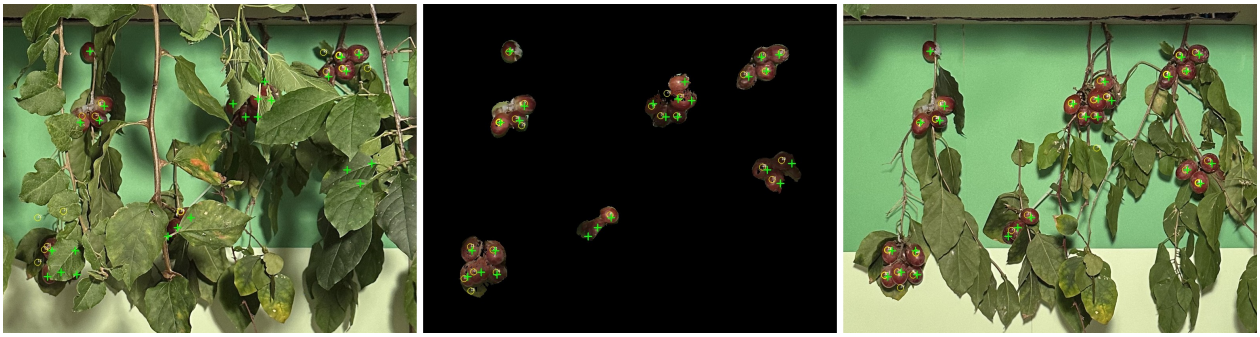


Figure 7. GrapeSAM detections (yellow circles) and hand labels (green crosses) for five branches at 20 mm: occluded reference (left), reconstruction (centre) and registered leaf-free photograph (right). The left and right panels show whole-image runs, so they include detections outside the fruit.

Table 2. GrapeSAM count F1 against 25 hand-labelled grapes. Single is the occluded view and Leaf free the registered photograph, both restricted to fruit pixels. Count is detections on the reconstruction.

Branches	mm	Single	Restored	Leaf free	Count
3	20	0.62	0.79	0.74	36
3	40	0.62	0.88	0.74	25
3	60	0.62	0.83	0.74	23
4	20	0.56	0.76	0.81	30
4	40	0.56	0.72	0.81	22
4	60	0.56	0.54	0.81	12
5	20	0.42	0.77	0.81	27
5	40	0.42	0.39	0.81	6
5	60	0.42	0.48	0.81	8
6	20	0.14	0.55	0.75	15
6	40	0.14	0.28	0.75	4
6	60	0.14	0.00	0.75	0

Table 3. Count F1 for three counters on 20 mm reconstructions. SAM 3 and Hough settings were selected on the four branch capture using the same labels, which favours them.

Branches	GrapeSAM	SAM 3	Hough
3	0.79	0.93	0.86
4	0.76	0.96	0.96
5	0.77	0.94	0.74
6	0.55	0.61	0.54

6.4. Computation

One sequence of 28 views took about 53 s end to end at scale 0.4 on a consumer PC, using stored masks. Scale 0.8 takes several minutes per capture, and SAM 3 mask generation is additional. This supports offline analysis, not real-time use on a vehicle.

7. Discussion

7.1. Why count on top of pixels

The two evaluations answer different questions. Pixel recall asks how much fruit area came back, while count asks whether that area is usable as fruit. They mostly agreed here, but count exposed differences pixels did not. At five branches and 20 mm, recall and F1 both improved substantially, yet at four branches and 60 mm a small recall gain (49% to 52%) came with no count benefit. Fragmented output can score on area while no longer being separable into grapes, so a count check is a useful addition to pixel evaluation for a crop-estimation goal.

7.2. What the experiments establish

With dense sampling, multiple views recovered hidden fruit and made it more countable by an independent model, consistent with the multi-view visibility effect of Hemming *et al*. [3] and the foreground suppression principle of Vaish *et al*. [7]. This is not an accuracy ranking against those studies, because their measures differ. Millan *et al*. [5] validate yield estimates, which this work does not. Closer spacing gave the largest gains under heavy occlusion, though it also changes view count, so the effect cannot be assigned to baseline alone.

7.3. Limitations

One scene. All conditions share one fruit arrangement and 25 grapes, so conditions are not independent, and there are no confidence intervals. **Dependence on SAM 3.** Fruit masks, foliage masks and truth masks all use SAM 3. GrapeSAM adds independence only at the counting stage. **Domain gap.** GrapeSAM was trained on vineyard images and its authors report weaker performance on indoor and uniform backgrounds [6]. Our composites, with blank backgrounds and cut edges, differ further, which may reduce or distort its scores. **Registration.** A single 2D transform per view treats fruit as one plane, which contributes to

offset in some clusters and to the truth comparison. **Generous matching.** The 24 pixel tolerance accepts off-centre detections. **Green grapes.** Green captures have no matching leaf-free photograph, so they were not evaluated here. In development, a green colour rule reached only IoU 0.31 against SAM 3 masks on a held-out capture, so colour alone looks unsuitable for them. **Laboratory motion.** A vehicle adds vibration, blur, speed changes, wind and changing light, none of which were tested.

No algorithm can recover fruit hidden from every view using direct visual evidence, and the output never fills such regions. A blank output must not be read as no grapes: masks, registration or view support may have failed, and a field system should report such cases as uninspectable.

7.4. Priorities for extension

The most valuable next step is repetition: more fruit arrangements with held-out labels, including green grapes with matching truth, to separate real effects from noise of the size seen here. Fine-tuning a counter is only justified after that, on labels not used for testing. Stabilising alignment under sparse views could reduce view loss at wider spacings. The deployment test should then mount one phone on a slow vehicle and measure overlap, blur and registration reliability under natural light, without external position or depth sensors. Ripe and unripe quantities, and calibration against harvested mass, remain further work.

8. Conclusion

The motivation is a low-complexity vineyard inspection system using a phone on an existing vehicle. Our controlled single-phone rail experiment supports one component of that concept: registering views on fruit and compositing only observed colours recovers hidden fruit, and that recovery can improve counting by an independent model. At 20 mm spacing with five branches, fruit pixel recall rose from 42% to 87% and count F1 from 0.42 to 0.77. Evaluating counts alongside pixels also showed where recovery was not useful, with benefits disappearing at wider spacing, and recovery only partial at six branches. These results come from one laboratory scene with 25 grapes, using SAM 3 masks, so field robustness, green grapes, ripe and unripe quantities and yield all remain unverified.

References

[1] Nicolas Carion, Laura Gustafson, Yuan-Ting Hu, Shoubhik Debnath, Ronghang Hu, et al. SAM 3: Segment anything with concepts. arXiv:2511.16719, 2025.

[2] Martin A. Fischler and Robert C. Bolles. Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981.

[3] Jochen Hemming, Jos Ruizendaal, Jan Willem Hofstee, and Eldert J. van Henten. Fruit detectability analysis for different camera positions in sweet-pepper. *Sensors*, 14(4):6032–6044, 2014.

[4] David G. Lowe. Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, 60(2):91–110, 2004.

[5] Borja Millan, Santiago Velasco-Forero, Arturo Aquino, and Javier Tardaguila. On-the-go grapevine yield estimation using image analysis and boolean model. *Journal of Sensors*, 2018: 9634752, 2018.

[6] Xiangzhi Tong, Chengrui Zhang, Mac Flaherty, Andre Matteo Garcia, Dominic Gorman, Jonathan Jaramillo, Justine E. Vanden Heuvel, and Yu Jiang. ViViD-5K: Vineyard vision dataset for field-based berry detection and segmentation and grape cluster closure estimation. arXiv:2605.24353, 2026.

[7] Vaibhav Vaish, Richard Szeliski, C. Lawrence Zitnick, Sing Bing Kang, and Marc Levoy. Reconstructing occluded surfaces using synthetic apertures: Stereo, focus and robust measures. In *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, pages 2331–2338, 2006.

[8] Yingqian Wang, Tianhao Wu, Jungang Yang, Longguang Wang, Wei An, and Yulan Guo. DeOccNet: Learning to see through foreground occlusions in light fields. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 118–127, 2020.

Statement of Contributions

Finn: acquisition, synthetic aperture and feature fusion (confirm scope).
Dinh: red/green segmentation and experimental results (confirm scope).
Has: related work, integration and conclusions (confirm scope).

Signatures: _____ _____ _____
Finn Dinh Has

Acknowledgment of AI Assistance

ChatGPT (OpenAI) and Claude (Anthropic) were used to assist with structuring and drafting the LaTeX report, clarifying methodological explanations, writing experiment and analysis code, generating results tables and figures, and checking citation and formatting consistency. The project team remains responsible for verifying all methods, measurements, citations, authorship and conclusions before submission.